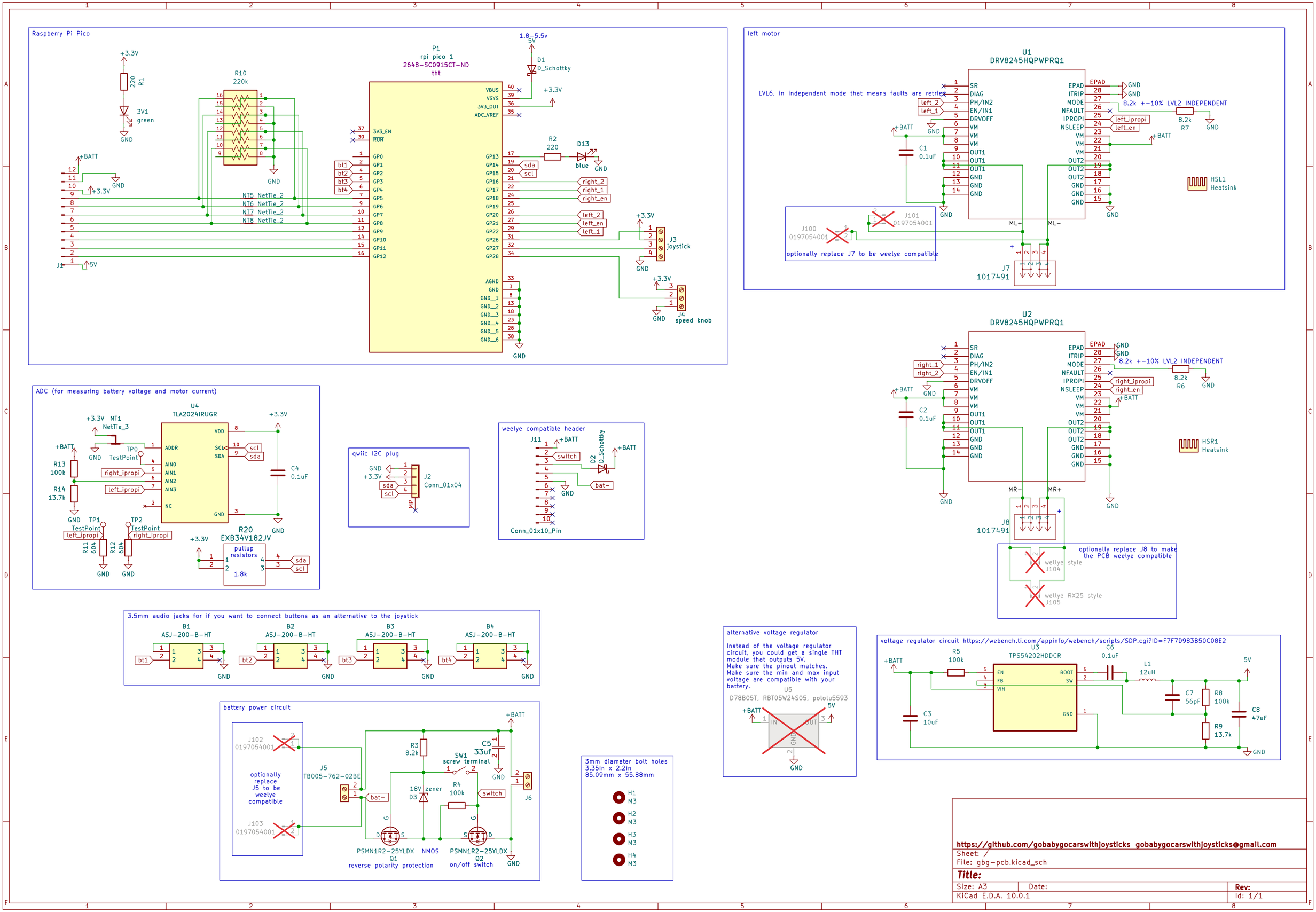

Raspberry Pi Pico
P1
rpi pico 1
2648-SC0915CT-ND
tht
R10
220k
R1
220
3V1
green
+3.3V
GND
+BATT
J1
5V
NT5 NetTie_2
NT6 NetTie_2
NT7 NetTie_2
NT8 NetTie_2
1.8-5.5v
5V
D1
D_Schottky
R2
220
D13
blue
sda
scl
right_2
right_1
right_en
left_2
left_en
left_1
J3
joystick
J4
speed knob
left motor
U1
DRV8245HQPWPRQ1
LVL6, in independent mode that means faults are retried
C1
0.1uF
8.2k +-10% LVL2 INDEPENDENT
R7
8.2k
left_ipropi
left_en
HSL1
Heatsink
J100
0197054001
J101
0197054001
optionally replace J7 to be weelye compatible
J7
1017491
ML+
ML-
U2
DRV8245HQPWPRQ1
C2
0.1uF
R6
8.2k
right_ipropi
right_en
HSR1
Heatsink
J8
1017491
MR-
MR+
optionally replace J8 to make the PCB weelye compatible
wellye style
J104
wellye RX25 style
J105
ADC (for measuring battery voltage and motor current)
U4
TLA2024IRUGR
NT1
NetTie_3
TP0
TestPoint
R13
100k
R14
13.7k
right_ipropi
left_ipropi
C4
0.1uF
TP1
TP2
R11
604
R12
604
R20
EXB34V182JV
pullup resistors
1.8k
qwiic I2C plug
J2
Conn_01x04
weelye compatible header
J11
switch
D2
D_Schottky
bat-
Conn_01x10_Pin
3.5mm audio jacks for if you want to connect buttons as an alternative to the joystick
B1
ASJ-200-B-HT
B2
ASJ-200-B-HT
B3
ASJ-200-B-HT
B4
ASJ-200-B-HT
battery power circuit
J102
0197054001
J103
0197054001
optionally replace J5 to be weelye compatible
J5
TB005-762-02BE
R3
8.2k
C5
33uf
SW1
screw terminal
18V zener
D3
R4
100k
J6
PSMN1R2-25YLDX
NMOS
Q1
reverse polarity protection
PSMN1R2-25YLDX
Q2
on/off switch
3mm diameter bolt holes
3.35in x 2.2in
85.09mm x 55.88mm
H1
M3
H2
M3
H3
M3
H4
M3
alternative voltage regulator
Instead of the voltage regulator circuit, you could get a single THT module that outputs 5V.
Make sure the pinout matches.
Make sure the min and max input voltage are compatible with your battery.
U5
D78B05T, RBT05W24S05, pololu5593
voltage regulator circuit https://webench.ti.com/appinfo/webench/scripts/SDP.cgi?ID=F7F7D983B50C08E2
U3
TPS54202HDDCR
R5
100k
C6
0.1uF
L1
12uH
C3
10uF
C7
56pF
R8
100k
R9
13.7k
C8
47uF
https://github.com/gobabygocarswithjoysticks gobabygocarswithjoysticks@gmail.com
Sheet: /
File: gbg-pcb.kicad_sch
Title:
Size: A3
Date:
Rev:
KiCad E.D.A. 10.0.1
Id: 1/1